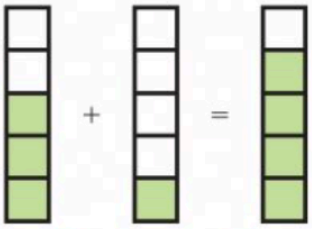
Exercise 3.3

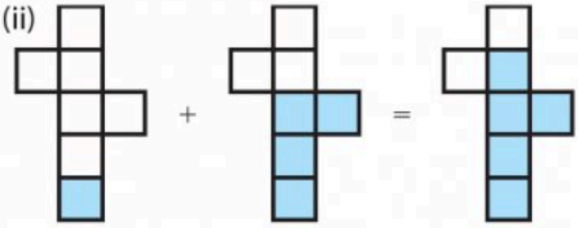
1. Copy and complete the fraction additions shown by these two diagrams: (no need to copy diagrams)

(i)



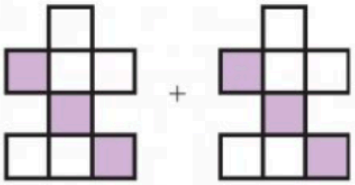
$\frac{\square}{5} + \frac{\square}{5} = \frac{\square}{5}$

(ii)



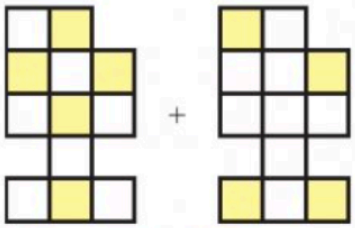
$\frac{\square}{7} + \frac{\square}{\square} = \frac{\square}{\square}$

(iii)



$\frac{\square}{\square} + \frac{\square}{\square} = \frac{\square}{\square} = \frac{\square}{4}$

(iv)



$\frac{\square}{\square} + \frac{\square}{\square} = \frac{\square}{\square} = \frac{3}{\square}$

2. Work out each of these and simplify your answer where possible.

(i) $\frac{2}{5} + \frac{1}{5}$ (ii) $\frac{3}{7} + \frac{2}{7}$ (iii) $\frac{1}{9} + \frac{2}{9} + \frac{4}{9}$

(iv) $\frac{4}{11} + \frac{2}{11} + \frac{1}{11}$ (v) $\frac{8}{9} - \frac{2}{9}$ (vi) $\frac{9}{10} - \frac{4}{10}$

(vii) $\frac{5}{7} + \frac{4}{7} - \frac{6}{7}$ (viii) $\frac{9}{13} - \frac{3}{13} + \frac{5}{13}$ (ix) $\frac{7}{8} + \frac{5}{8} - \frac{3}{8}$

3. Simplify these calculations as far as possible:

(i) $\frac{1}{4} + \frac{3}{8}$ (ii) $\frac{3}{10} + \frac{2}{5}$ (iii) $\frac{5}{6} + \frac{2}{3}$

(iv) $\frac{3}{7} + \frac{1}{14}$ (v) $\frac{5}{12} + \frac{3}{4}$ (vi) $\frac{2}{5} - \frac{1}{15}$

(vii) $\frac{2}{3} - \frac{1}{4}$ (viii) $\frac{9}{14} - \frac{1}{7}$ (ix) $\frac{11}{20} - \frac{1}{5}$

4. Work out each of the following and simplify your answers:

(i) $\frac{1}{3} + \frac{2}{5} + \frac{1}{15}$ (ii) $\frac{3}{8} + \frac{1}{6} + \frac{2}{3}$ (iii) $\frac{5}{12} + \frac{3}{4} + \frac{1}{3}$

(iv) $\frac{2}{9} + \frac{1}{3} + \frac{1}{6}$ (v) $\frac{3}{4} + \frac{1}{2} - \frac{1}{3}$ (vi) $\frac{7}{12} + \frac{1}{8} - \frac{1}{6}$

5. Work these out and simplify your answers:

(i) $2\frac{1}{3} + 1\frac{1}{6}$ (ii) $2\frac{1}{4} + 1\frac{1}{3}$ (iii) $3\frac{1}{2} + 2\frac{3}{5}$

(iv) $3\frac{3}{10} + 1\frac{4}{5}$ (v) $1\frac{3}{4} + 2\frac{5}{12}$ (vi) $4\frac{1}{5} + 1\frac{1}{3}$

(vii) $2\frac{7}{10} + 1\frac{3}{5}$ (viii) $3\frac{1}{5} + 2\frac{2}{3}$ (ix) $3\frac{2}{3} - 2\frac{1}{2}$

6. Match each calculation with its result.

A	$\frac{3}{4} - \frac{1}{4}$	B	$\frac{2}{9} + \frac{1}{9}$	C	$\frac{3}{4} + \frac{3}{4}$	P	$\frac{1}{3}$	Q	$\frac{2}{5}$	R	$\frac{1}{2}$
D	$\frac{3}{10} + \frac{1}{10}$	E	$3\frac{3}{8} - 2\frac{5}{8}$	F	$1\frac{5}{6} - 1\frac{1}{6}$	S	$1\frac{1}{2}$	T	$\frac{2}{3}$	U	$\frac{3}{4}$

7. Work out each of these and verify your answers by using your calculator.

(i) $\frac{7}{8} + \frac{3}{4}$ (ii) $\frac{5}{6} + \frac{7}{12}$ (iii) $\frac{11}{15} - \frac{2}{5}$

(iv) $4\frac{3}{4} - 2\frac{1}{2}$ (v) $2\frac{7}{8} - 1\frac{3}{4}$ (vi) $5\frac{2}{3} - 3\frac{1}{4}$

8. Work out each of these and verify your answers by using a calculator.

(i) $5\frac{3}{4} - 2\frac{1}{3}$ (ii) $4\frac{3}{10} - 2\frac{2}{5}$ (iii) $5\frac{2}{9} + 3\frac{1}{3}$

(iv) $4\frac{3}{8} + 2\frac{1}{2} + 1\frac{3}{4}$ (v) $2\frac{7}{10} + 2\frac{3}{5} - 3\frac{1}{2}$ (vi) $3\frac{5}{8} + 2\frac{3}{4} - 4\frac{1}{6}$

9. In a magic square, each row, each column and each diagonal add up to the same number. Copy and complete this magic square.

$5\frac{1}{3}$		
$\frac{2}{3}$	$3\frac{1}{3}$	6